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Framework for Rail Transport Inequality Assessment: A Case Study of the Indian Railway Zones with Superfast Express (SE) Trains

Shailesh Chandra * and Vivek Mishra

Department of Civil Engineering and Construction Engineering Management, California State University, Long Beach, CA 90840, USA

* Correspondence: shailesh.chandra@csulb.edu

Abstract: The paper presents a framework for assessing inequality in passenger rail services by examining connectivity and accessibility across two different travel frames: favorable (daytime travel) and unfavorable (nighttime travel). The unfavorable timeframe is often characterized by a lack of adequate first-/last-mile transport availability to train stations, impacting passenger access. The study employs a newly developed inequality-to-investment ratio to classify rail transport entities like stations, routes, or zones as either winners or losers, using a performance persistence analysis. A case study was conducted using data on the superfast express (SE) train schedule and investments from 2016 to 2020 across sixteen zones of the Indian Railway, assessing service inequalities during the hours of favorable (7 a.m. to 9 p.m.) and unfavorable (9 p.m. to 7 a.m.) timeframes. Results revealed inconsistencies in inequalities among zones. Moreover, the research demonstrates that the choice between connectivity and accessibility metrics significantly influences the identification of better or worse performing zones as winners or losers, respectively, throughout the 2016–2020 period and during both timeframes. The study underscores the importance of selecting appropriate measures and suggests re-evaluating or increasing investments in certain zones of Indian Railways based on these insights. The framework proposed in this paper can be used to assess inequalities within any transportation system receiving investments or government funds, contributing to the broader goal of ensuring equitable access to transportation, which is essential for a sustainable transportation system.



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Keywords: passenger rail; superfast express trains; connectivity; accessibility; inequality; Indian Railway; sustainable transport

1. Introduction

Passenger rail is critical to the economic growth of both developed and developing economies [1–3]. For example, in India, the intercity passenger rail is vital to transportation needs, having large-scale impacts on the country's economy [4]. In other countries like China, connectivity by train is so vital that 19 of the 20 top-ranking domestic transportation routes are by rail [5,6]. The dependency on rail in other countries like the USA is similar, which is critical in its economic sustenance [7].

Passenger rail facilitates access to opportunities, and the lack of it and, in general, any public transportation facility translates to social exclusion emerging due to a lack of social opportunities [8]. Several scholars agree that rail's influence on the economy rests on its ability to address inequalities, fulfilling the need to provide a means of transport for justice and fairness between cities experiencing differential economic activities [9,10]. Thus, in essence, the role of rail in the economic development of a nation is evident in its meeting the equity goals of connecting places and attenuating regional disparities of spatially far-apart places [11–13].

Some studies, however, have highlighted the disadvantages of rail access [14–17]. They have noted that rail transportation has caused wealthy cities to enjoy higher accessibility benefits than hinterland cities [18]. Similar observations were made for cities with rail transportation in Europe [19].

Inequalities stemming from transit connectivity and accessibility are contingent upon essential factors and amenities in the vicinity of the stations utilized by train passengers. The popularity and ridership of a rail system hinge on the effective connection of its stations with first-/last-mile transportation. This connection significantly contributes to the overall service quality, which is particularly important for low-income shift workers, especially those commuting during nighttime when first-/last-mile connectivity is often limited [20]. Research shows discomfort and insecurities arise in accessing transit stations during often unconducive or odd times of the day. For example, Jehle et al. [21] have pointed out the challenges passengers often face in being unable to access railway stations due to a lack of safety or first-/last-mile connectivity at night. Other scholars have also emphasized these findings on the need for nighttime transport, whether in a developed [22] or a developing nation [23].

During nighttime or unconventional hours of travel, passengers may face safety challenges at a station due to insufficient infrastructure facilities, such as inadequate street lighting and patrolling. These hours are particularly ‘unfavorable’ for vulnerable populations undertaking travel [24]. Lack of first-/last-mile transport connectivity at night signifies increased inequalities [25]. Hence, it is imperative to implement policies that mitigate inequality, with a specific focus on enhancing equity in rail services. This approach is essential to ensure fair access to resources, a prerequisite for both economic growth and upliftment. To formulate effective policies, it is crucial to first understand the extent of existing inequalities.

Often, investments (especially capital) in rail are made to eliminate regional inequalities and provide equal opportunities for transport access to all. However, this has not always been the case. Constantin et al. [17] showed that despite substantial investments, the Romanian Railway did not positively contribute to reducing regional inequalities. This occurred due to uneven growth and low accessibility.

In general, evaluating the effects of inequality can be a complex task in most research findings related to rail connectivity and accessibility. Recommendations have been proposed to employ spatial impact analysis based on changes in accessibility distribution within the particular urban clusters affected by rail projects [10]. Proximity to population centers, the presence of rail stations, and the service quality of the overall transportation system impact the inequality implications [20], which not many studies have addressed comprehensively in the past for semi-high-speed, premium, or superfast rail transportation in developing countries.

Therefore, this research is focused on evaluating inequalities among service zones of rail on its performance measures of connectivity and accessibility. Further, this paper enriches the knowledge of the performance of entities such as zones or regions of operations of rail services, identifying them as either beneficiaries (as winners) or disadvantaged (as losers), focusing on the investment patterns and shifts in inequality over time.

The framework presented in this paper is specifically studied with an application for India, a developing nation, with the impacts of the superfast express (SE) trains of the Indian Railway. The SE train impacts are studied on the inequalities of the sixteen zones of the railways utilizing data on train arrivals and departures from 2016 to 2020. The results are evaluated for two passenger travel timeframes: (i) between 7 a.m. and 9 p.m. (during daytime and favorable), and (ii) between 9 p.m. and (next day) 7 a.m. during nighttime as unfavorable. While other boundary times could have been chosen, our focus was to identify when first-/last-mile connectivity to stations is typically at its lowest in a developing country, which usually occurs after dark. We assumed this period to be from 9 p.m. to 7 a.m. Therefore, the 9 p.m.–7 a.m. timeframe was considered ‘unfavorable’ for passengers (in contrast to the 7 a.m.–9 p.m. timeframe, which was considered favorable),

mainly due to the presumed lack of first-/last-mile connectivity and safety concerns during nighttime hours. These timeframes were selected to highlight the differences in inequalities between daytime and nighttime travel. The findings shed light on any policy-related interventions that might be needed to minimize the inequalities that were observed, which can be brought about by reviewing existing train schedules, ensuring first-/last-mile access to stations at night, etc.

2. Research Contributions

Although the importance of passenger rail in achieving equity goals has been frequently highlighted in the literature, evaluating its impact due to investments in transport infrastructure remains complex. Past studies on economic agglomeration at a spatial level indicate that increased infrastructure investments and connectivity can exacerbate regional disparities. This occurs when larger cities draw more resources from newly connected smaller cities [12]. Connectivity between wealthy and poor regions does not always yield mutual benefits and can result in a ‘siphon effect’ where resources flow from small cities to large cities due to low transportation costs [26,27]. Therefore, public spending on rail transportation infrastructure aimed at benefiting all regions, especially those needing development, might backfire and lead to further marginalization due to this siphoning effect. Our research contributes to the literature on inequality and passenger rail transportation in the following four ways:

First, given the challenge of a potential siphoning effect, it is crucial to investigate to what extent passenger rail can meet equitable connectivity and accessibility goals for regions that consistently receive public funding. Our study complements the efforts of other studies that emphasize addressing transportation inequality, which, if unmet, can hinder sustainable economic growth in a region. Numerous proponents stress that integrating equity into transportation decision-making should be a priority for public agencies and policymakers [28,29]. There are multiple instances where public agencies have sought to reduce inequality by considering it a performance measure for transportation infrastructure investments, both in developed nations like the United States [30] and in developing countries such as India [22] and China [31]. Our research provides a framework to assess how effectively public investments in passenger rail can fulfill inequality goals when these efforts are prioritized.

Second, our research framework and methodology can inform regional planning initiatives by identifying areas needing focused attention through targeted transportation interventions. This will support the inequality efforts of passenger rail systems globally. For instance, this is relevant for Europe as it strives to meet its 2030 and 2050 plans and investments in the Trans-European Transport Network (TEN-T), which is proposed to ensure connectivity and accessibility of the European Railway network. Similarly, the framework for evaluation presented in this paper could be applicable to India, as the Indian Railways plans to invest USD 715 billion by 2030 in the passenger rail infrastructure [31]. Thus, our research can help ensure that these investments are effective in addressing regional disparities and achieving equitable outcomes.

Third, as we have found in our extensive review of the literature, none of the prominent studies have successfully integrated both connectivity- and accessibility-based formulations into inequality assessments that are directly linked to investments. For instance, Liu et al. [26] focus solely on incorporating centrality into inequality formulations, and Zhou et al. [32] employ a connectivity–accessibility index followed by a performance persistence analysis. However, both studies fall short in incorporating investment data or other investment proxies to thoroughly explore the relationship between institutional efforts (investments) and their outcomes (inequality). Our research stands out with a unique contribution: a newly developed inequality-to-investment ratio, complemented by a robust performance persistence analysis to identify and test high-performing (winner) and low-performing (loser) zones of interest for consistency.

Fourth, investments in rail infrastructure and operations often do not seem to adequately address inequalities. It remains uncertain whether past or future investments will effectively bridge the gap between high-performing and lower-performing operational regions of the rail over a period. This inquiry aligns with the broader goal of public transit systems, which is to understand the impact of connectivity and accessibility between different entities over time, whether they are cities [18] or zones. We showcase our framework's capability to link investments with inequality that could yield winners and losers through the example of the superfast express (SE) trains of the Indian Railways.

3. Materials and Methods

We first adopt a conventional formulation for connectivity and accessibility (two important performance measures in transit studies) by incorporating them into the inequality formulation for evaluating the performance of a typical passenger rail system. Centrality, a key topological feature in network analysis, is employed for the connectivity aspect, a concept extensively utilized in various fields like physical networks, computer science, and epidemiology [33]. While connectivity captures a network's topological characteristics, accessibility serves as another frequently employed metric for assessing passenger flow [26].

Using the two measures of connectivity and accessibility, a network's two important features, namely, the topology and the population (as a surrogate of passenger flow), were simultaneously considered. Note that past studies have carried out similar accessibility calculations with population data surrounding stations or stops rather than considering actual passenger flow or ridership [34,35]. Though various forms of these two measures exist in the literature, none we came across were sensitive to the train's scheduled arrival or departure times at a station within a network. Therefore, both connectivity and accessibility measures were first developed to be time-sensitive, reflecting the train's schedule effects.

3.1. Connectivity

The 'degree centrality' formula has been modified to develop the connectivity measure. The degree centrality of a node is the number of edges it has [36]. It is usually deployed to identify the most important node in a network and indicates the number of neighboring stations directly close to a station [37]. This means that the higher the centrality of a node, the more central and connected it is to the other nodes. In the formulation for centrality presented here, we use the centrality measure as a station's connectivity, which should be sensitive to the favorable (or unfavorable) travel times. Our connectivity formula uses the number of trains instead of the number of edges, with each train arriving at or departing from a station representing an edge.

The connectivity formula (denoted using $C_{i,z}^y$) for a node or a station i in zone z in the year y is expressed as follows:

$$C_{i,z}^y = \sum_{k=1}^K \left(\sum_{i \neq j \in N} \tau_{j,i,a}^{k,z,y} w_{ji}^{k,z,y} + \sum_{i \neq j \in N} \tau_{i,j,d}^{k,z,y} w_{ij}^{k,z,y} \right) \quad (1)$$

where

i, j = a station in a zone;

k = a train running between stations;

y = year of analysis;

z = a zone with stations;

a = train arrival;

d = train departure;

K = the number of trains;

N = number of stations being considered in the network;

$\tau_{j,i,a}^{k,z,y} = 1$ if a train k departs station j to arrive at station i (in zone z) during favorable travel timeframe in a given year y ; otherwise, it is 0 for unfavorable timeframe;

$\tau_{i,j,d}^{k,z,y} = 1$ when in a given year y a train k departs station i (in zone z) to arrive at station j during favorable travel timeframe; otherwise, it is 0 for unfavorable timeframe;
 $w_{ij}^{k,z,y} = 1$ if in a given year y there exists a train k from station i (in zone z) to station j , 0 otherwise;
 $w_{ji}^{k,z,y} = 1$, if there exists a train k from station j to station i (in zone z), 0 otherwise.

3.2. Accessibility

The accessibility used in the paper is a potential accessibility measure that has found a wide use and strong representation in the economic geography of a region, as evidenced by its use in various studies by Rotoli et al. [38] and Zhou et al. [32]. The potential accessibility measure is the state-of-the-art application of an empirically constructed measure and represents a direct relationship between population (often called ‘opportunity’) and is inversely proportional to the impedance between studied stations or zones. Accessibility derived from a transportation facility (such as rail) is often used as a representation of a region’s economic potential and attractiveness [39].

With a focus on the accessibility (or attraction) of a station in a zone, a formula is proposed for accessibility using a dummy variable $\delta_{i,j}^k$ and considering the year of analysis. The accessibility $A_{i,z}^y$ of station i in zone z in the year y is defined as follows:

$$A_{i,z}^y = \sum_{k=1}^K \sum_{j=1}^N \frac{\delta_{i,j,k}^y P_j^y}{I_{i,j,k}^y} \quad (2)$$

where

$\delta_{i,j,k}^y = 1$ if in a given year y , the arrival time of train k from station i (in zone z) to station j is within the favorable travel timeframe; otherwise, it is 0 if it is to be in the unfavorable travel timeframe;

P_j^y = the population of the city/town with station j for the year y ;

$I_{i,j,k}^y$ = the impedance in the year y between station i and station j with train $k \in K$ (total number of trains). The impedance used is the travel time between stations, with additional elaboration on its application as demonstrated in the case study;

N = the number of stations in the network.

The other common variables are defined as noted under Equation (1).

3.3. Generalized Connectivity–Accessibility Index

For a passenger, although a station with the largest connectivity would be attractive, there is a possibility that this may not be true when travel times from that station to the other stations are high since connectivity expressed in Equation (1) does not take into consideration the travel time between stations. On the other hand, accessibility—involving station/city population and travel times between stations—presents a more holistic measure of the attractiveness for rail passengers [39].

As evidenced by research, both connectivity and accessibility are equally important for enjoying the benefits of rail transportation [32]. To account for both centrality or connectivity and accessibility measures, we develop a generalized measure ($\Omega_{i,z}^y$) for a station i in zone z for the year y , expressed as follows:

$$\Omega_{i,z}^y = \begin{cases} C_{i,z}^y & \text{if connectivity is sought} \\ A_{i,z}^y & \text{if accessibility is sought} \end{cases} \quad (3)$$

Therefore, the generalized connectivity and accessibility index, Ω_z^y , for a zone (with stations) is the average of $\Omega_{i,z}^y$ of all the stations contained in that zone.

3.4. Inequality Measurement

Public transportation projects should consider equity during the planning stages, and appropriate inequality measuring instruments must be used [40]. In this paper, the inequality is measured at the regional or zonal level based on the connectivity and accessibility of all the stations in a zone. Analyzing inequality or disparity among zones at the spatial scale can reveal gaps in rail connectivity and accessibility, which can further justify and guide future zonal-level investments. Thus, historically, zones with stations that have benefited or not benefited from train service during favorable (or unfavorable) travel timeframes are identified with the disparity analysis. A disparity analysis can reflect any modifications, expansions, or adjustments needed in operating the train and expanding service connectivity among the cities within regions or zones.

In this paper, we use a modified form of the entropy index (Theil's T index) as a decomposable measure of inequality that can be disintegrated into population groups, income sources, or other dimensions [41,42]. Although there are some commonly available inequality measuring indicators (such as Gini and Lorenz curves) that are widely used in policy-related issues in transportation, the use of Theil's index is appropriate for our case as the index is much sensitive (due to its logarithmic structure) to even small variations in connectivity (and accessibility) of several regions or zones such as deployed for the case study later on the Indian Railways. While the use of other indices would also be appropriate—indeed, the Gini index produced relatively similar results for the zones—our decision to use Theil's index was driven by the need to enhance the sensitivity of the inequality outputs, particularly when two or more zones exhibited inequalities that were numerically very close.

In practice, Theil's index has been successfully used for measuring accessibility-based inequalities [43] and for measuring regional inequalities [26,44].

The modified Theil T index (T_z^y) for a zone z in year y incorporating the generalized connectivity–accessibility index of a station is expressed as follows:

$$T_z^y = \frac{1}{n_z} \sum_{i=1}^{n_z} \frac{\Omega_{i,z}^y}{\Omega_z^y} \ln \left(\frac{\Omega_{i,z}^y}{\Omega_z^y} \right) \quad (4)$$

where

n_z = number of stations in a zone z (assumption is that the total number of stations remains constant during each analyzed year y);

$\Omega_{i,z}^y$ = measure of the connectivity or accessibility of station i in zone z in year y ;

Ω_z^y = mean connectivity or accessibility for all the stations in zone z during year y .

We use the inequality formulation above to evaluate the performance of zones for connectivity and accessibility, as described in the next section.

3.5. Performance Persistence Analysis

We utilize a nonparametric method to conduct a performance assessment of a zone for inequality with connectivity and accessibility over a specified investment period. The performance analysis guides stakeholders to achieve regional and social inequality by directing investments in public transport [45–47]. The performance persistence analysis is an established approach that has been used to track the evolution of winner and loser status of analyzed entities such as zones, allowing for the evaluation of the relationship between past and future performances [33]. Zonal-level performance analysis should be conducted to understand how zones have improved in inequality over the years through policy interventions, particularly from investments. Therefore, we classify zones into winners and losers based on their performance, assessed through inequality improvement for the investment they received.

Utilizing the fundamentals of the performance persistence analysis by Agarwal and Naik [48], an inequality-to-investment ratio is developed. The inequality used here can be derived interchangeably from connectivity or accessibility using Equations (3) and (4).

The inequality-to-investment ratio ($E_z^{y-1,y}$) is defined as the percentage change in inequality of a zone z divided by the percentage change in investment from year $y - 1$ to y and is expressed as,

$$E_z^{y-1,y} = \left(\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}} \right) / \left(\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}} \right) \quad (5)$$

where I_z^y is the investment in zone z at the beginning of fiscal year y and T_z^y is the inequality at the end of fiscal year y .

The purpose of our assumption in providing linearly comparable improvements due to investment in Equation (5) is to simplify the decision-making process for public transport managers and stakeholders, enabling them to gather insights quickly without needing to calibrate parameters that would arise from a complex model. Such a model would precisely describe the relationship between zone inequality and investments.

The assumption with the ratio $E_z^{y-1,y}$ is that it reflects the investments at the beginning of a fiscal year y that bring about changes for inequality (which is connectivity- and accessibility-based) during the year y itself. This suggests the effectiveness of promptly utilizing rail investments to reduce disparities in specified regions or study zones.

Decision-makers typically finalize significant monetary investments in infrastructure within a zone well in advance of the start of a fiscal year. If, at the conclusion of the fiscal year, the inequality in that zone remains notably high despite the investment, it is regarded as an ineffective allocation of funds, especially in zones where reducing inequality poses significant challenges. Hence, it is crucial to evaluate zones that receive investments by identifying those that succeed and those that struggle based on this principle. We, therefore, classify zones as winners (that are high-performing and succeed) and loser zones (that fail or struggle) based on this evaluation scheme utilizing the proposed ratio $E_z^{y-1,y}$ in Equation (5) and as further illustrated in the next section.

Determining Winners and Losers

We assume that there must have been a direct or an indirect impact of investment in causing a decrease or an increase in inequality. Although it can be noted that other causal factors might impact the inequality and investment relationship, we excluded this investigation since it will digress from our current focus.

As evident from the formulation of the ratio $E_z^{y-1,y}$ in Equation (5), two cases can arise, as described below:

The first case occurs when $E_z^{y-1,y}$ is positive, which can happen under two scenarios:

1. When both $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ and $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$ are positive. This implies that $I_z^y > I_z^{y-1}$ and $T_z^{y-1} > T_z^y$, indicating that an increase in investment has occurred while inequality has decreased from $y - 1$ to y , which is desirable by stakeholders and investors;
2. When both $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ and $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$ are negative, meaning $I_z^y < I_z^{y-1}$ and $T_z^{y-1} < T_z^y$. This reflects a decrease in investment from $y - 1$ to y , which led to an increase in inequality, an undesirable outcome.

The second case is when $E_z^{y-1,y}$ is negative for a zone z . This can happen under two conditions:

1. $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ is positive and $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$ is negative, indicating that investment increased from year $y - 1$ to y ($I_z^y > I_z^{y-1}$) but inequality also increased ($T_z^{y-1} < T_z^y$), which is undesirable since investment is expected to reduce inequality;
2. $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ is negative while $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$ is positive, meaning investment decreased from $y - 1$ to y ($I_z^y < I_z^{y-1}$), yet inequality also decreased ($T_z^{y-1} > T_z^y$), an unusual outcome as reduced investment typically leads to increased inequality.

A zone is considered a ‘winner’ when the ratio $E_z^{y-1,y}$ is positive, based on the values of $\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}}$ and $\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}}$, as indicated by the green-shaded cells in Table 1. A winner zone is characterized by $E_z^{y-1,y} > 0$ (i.e., $T_z^{y-1} > T_z^y$ and $I_z^y > I_z^{y-1}$). Conversely, the red-shaded cells represent loser zones, where inequality increased despite changes in investment from $y - 1$ to y . A loser zone z is defined by $E_z^{y-1,y} < 0$ with $I_z^y > I_z^{y-1}$ and $T_z^{y-1} < T_z^y$ or $I_z^y < I_z^{y-1}$ and $T_z^{y-1} > T_z^y$.

Table 1. Scenarios for winner or loser zones (green shade indicating winner, red shade indicating loser, and gray shade indicating neutral).

	$\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}} > 0$	$\frac{I_z^y - I_z^{y-1}}{I_z^{y-1}} < 0$
$\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}} > 0$	(+)	
$\frac{T_z^{y-1} - T_z^y}{T_z^{y-1}} < 0$	(−)	(+)

In the cases presented above, a zone experiencing an increase in inequality or decrease in investment from $y - 1$ to y would indicate diminished connectivity and accessibility.

The grayed-out cell in Table 1 indicates an anomaly, as the investment has decreased from year $y - 1$ to y but has caused a decrease in inequality for the same period. Any zone falling in this classification is a neutral zone.

Once the winner/loser zones are determined, the cross-product ratio is computed using the odds ratio of the number of repeat performers, whether winners or losers, in two consecutive years. The formula for the odds ratio, $O_{y-1,y}$, is expressed as follows:

$$O_{y-1,y} = \left(\frac{WW_{y-1,y} \times LL_{y-1,y}}{WL_{y-1,y} \times LW_{y-1,y}} \right) \quad (6)$$

where

$WW_{y-1,y}$ = number of zones that are categorized as winners for two consecutive years ($y - 1$ and y);

$LL_{y-1,y}$ = number of zones that are categorized as losers for two consecutive years ($y - 1$ and y);

$WL_{y-1,y}$ = number of zones that are categorized as winners in year $y - 1$ and losers in the year y ;

$LW_{y-1,y}$ = number of zones that are categorized as losers in $y - 1$ and winners in y .

An odds ratio above 1 indicates a positive association between an outcome (e.g., being a winner or loser) in one time period and the same outcome in the subsequent period. On the other hand, an odds ratio of less than 1 indicates a negative association, which means that the outcome in the following period is more likely to be the opposite of the outcome in the previous period.

3.6. Example Calculation

A summarized version of the proposed framework is presented in the flowchart shown in Figure 1 below. Additionally, an example calculation with assumed data for two zone examples is provided in Figure 2, illustrating the steps for implementing the inequality assessment to identify winner and loser zones.

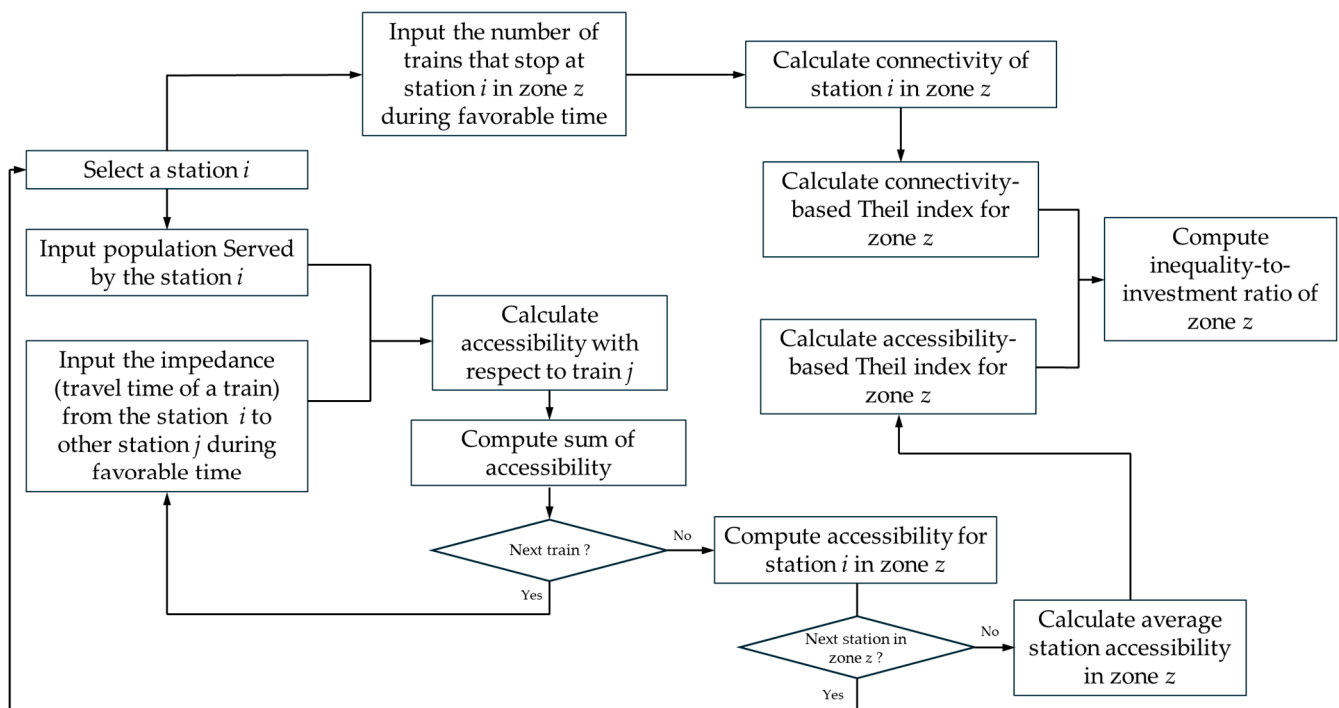


Figure 1. Flowchart for inequality calculation of a zone.

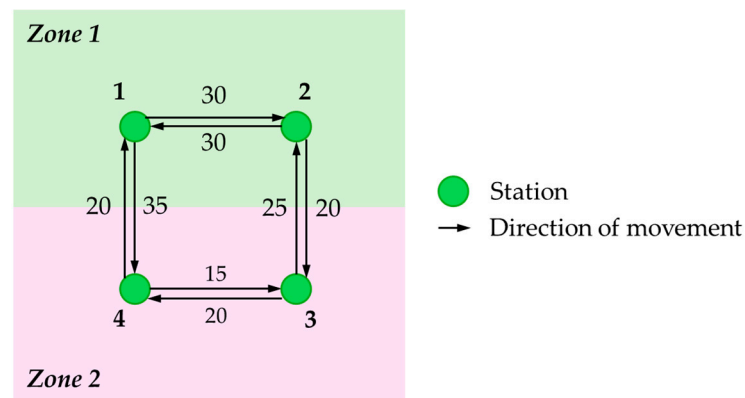


Figure 2. Setup example of inequality calculation with four stations and two zones.

Referring to the setup in Figure 2, with impedance (travel time) indicated for each direction of movement, let us consider data from two years: Year 1 and Year 2. For zones 1 and 2 during these two years, the data required to determine the winner and loser zones are as follows:

- Origin–destination travel time data in Table 2, as referenced in Figure 2;
- Frequency of trains arriving and departing from each station in Table 3;
- Population served by each station in Table 4.

Table 2. Origin–destination (OD) travel time between stations for Year 1 and Year 2.

Station	1	2	3	4
1		30	50	35
2	30		20	40
3	40	25		20
4	20	40	15	

Table 3. Frequency of trains arriving and departing for Year 1 (with data for Year 2 presented in parentheses).

Station	1	2	3	4
1		2 (3)	3 (5)	4 (6)
2	3 (5)		4 (6)	5 (7)
3	4 (6)	5 (7)		6 (8)
4	7 (9)	8 (10)	9 (11)	

Table 4. Population served by stations of trains for Year 1 and Year 2.

Station	1	2	3	4
1		2000	2000	2000
2	3000		3000	3000
3	3000	3000		3000
4	4000	4000	4000	

The subsequent calculations for the connectivity, accessibility, and inequality for the two years for the two zones are shown in Tables 5–7.

Table 5. Calculation output for Year 1.

Station	Connectivity	Normalized Connectivity	Mean Connectivity of Zone	Theil's Index with Connectivity	Accessibility	Normalized Accessibility	Mean Accessibility of Zone	Theil's Index with Accessibility
1	23	0.19			424	0.17		
2	27	0.23	0.21 (Zone 1)	0.0032	470	0.19	0.18 (Zone 1)	0.0014
3	31	0.26			732	0.30		
4	39	0.33	0.29 (Zone 2)	0.0065	795	0.33	0.32 (Zone 2)	0.0009

Table 6. Calculation output for Year 2.

Station	Connectivity	Normalized Connectivity	Mean Connectivity of Zone	Theil's Index with Connectivity	Accessibility	Normalized Accessibility	Mean Accessibility of Zone	Theil's Index with Accessibility
1	34	0.20			539	0.19		
2	38	0.23	0.22 (Zone 1)	0.0015	612	0.22	0.21 (Zone 1)	0.0020
3	43	0.26			802	0.29		
4	51	0.31	0.28 (Zone 2)	0.0036	849	0.30	0.29 (Zone 2)	0.0004

Table 7. Output with assumed investment data for zones.

	Investment (Assumed in Crore INR)		Ratio of Change in Investment	Ratio of Change in Theil's Index (Connectivity)	Ratio of Change in Theil's Index (Accessibility)	Inequality-to- Investment Ratio (Connectivity)	Inequality-to- Investment Ratio (Accessibility)
	Year 1	Year 2					
Zone 1	100	300	$(300 - 100)/100 = 2$	$(0.0032 - 0.0015)/0.0032 = 0.52$	$(0.0014 - 0.0020)/0.0014 = -0.49$	$=0.52/2 = 0.26$ (Winner)	$-0.49/2 = -0.25$ (Loser)
Zone 2	200	400	$(400 - 200)/200 = 1$	$(0.0065 - 0.0036)/0.0065 = 0.45$	$(0.0009 - 0.0004)/0.0009 = 0.53$	$=0.45/1 = 0.45$ (Winner)	$-0.53/1 = 0.25$ (Winner)

Thus, for the example shown above, although zone 1 turns out to be a winner for the connectivity measure, it is a loser for the accessibility measure. Zone 2 proves to be a winner for both measures.

4. Application: A Case Study with SE Trains in India

The passenger rail system in India is a critical component of the country's transportation infrastructure, playing a pivotal role in connecting diverse regions and fostering economic development. India's railway network has undergone significant expansion

across the nation, boasting a total route length of 67,415 km in 2019. In the fiscal year 2018–19, the Indian Railways documented nearly 8.5 billion originating passengers, signifying individuals whose journeys commenced with the train. The passenger-kilometers for suburban and suburban regions were approximately 1.2 trillion during this period [49].

Our focus in this case study is particularly on the superfast express (SE) trains of the Indian Railways. The SE trains are the long-distance premium services offered by the Indian Railways, known for higher speed and better amenities. These trains constitute almost 10% of the total trains (both suburban and non-suburban) operated by the Indian Railways [50,51] and with additional superfast charges levied over regular mail/express train charges (etrain.info, 2021 [52]). Further, these trains make fewer stops compared to their counterparts (i.e., other mail/express trains), allowing them to cover the same distance in a significantly shorter time span. However, there is a growing concern about the potential inequality in the distribution and accessibility of these SE trains and the services they provide across various zones of the Indian Railways [53].

Presently, India manages an extensive network of around 1200 SE trains, which includes renowned services such as Rajdhani Express, Durgam Express, Shatabdi Express, and various other superfast trains (etrain.info [52]). As reported by Indian Railways [12], these trains consistently maintain average speeds ranging from 85 to 95 km/hr. The average distance of travel by various SE trains ranges from 290 km to 1930 km [54] and has impacts on the sixteen zones of the Indian Railways, as indicated on the map presented in Figure 3. Note that the seventeenth zone, the Konkan Railway, was left out of this analysis as there were no available data on investment by the Indian Railways for the years of analysis.

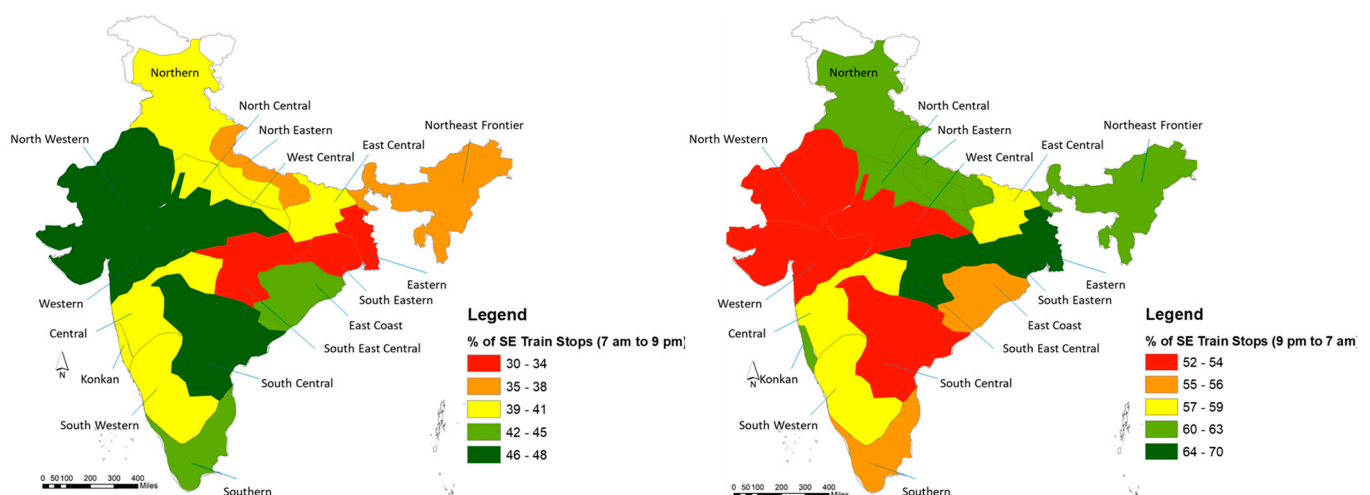


Figure 3. Percentage of SE train stops (at stations) across the zones.

Additionally, the three zones of Eastern, Southeastern, and Southeast Central were found to have the largest percentage (about 64–70%) of SE train stops during the 9 p.m.–7 a.m. timeframe. The Western, Northwestern, and South Central zones have the lowest percentage of about 52–54% SE train stops during the same travel period. Other details for all other zones are presented in the map of Figure 3.

The zones represent economically significant areas of operation for the Indian Railways, with SE trains and the network connecting the diverse socio-economic landscapes of the country within these seventeen zones. The operations of the Indian Railways in these zones continue seamlessly day and night [55]. However, our focus is only on those stations that are stops of SE trains. Thus, we study the impacts of inequality using the methodology presented in the earlier sections of this paper only for these stations for the two timeframes: 7 a.m.–9 p.m. (daytime as favorable) and 9 p.m.–7 a.m. (nighttime as unfavorable).

The subsequent section details the data collection efforts, some key findings, and outlines the assumptions made in this case study.

Data Collection: Travel Time, Population, and Investment

The spatial data of the SE train stations (as stops) and the rail network of the Indian Railways were obtained from various online sources (including [Mapcruzin.com](https://www.mapcruzin.com) [56,57]). The analysis was conducted with SE train arrival/departure data at 1620 stations with 1200 SE trains running per year (whether daily, weekly, or otherwise) and collected from 2016 to 2020. The stations served as a stop to at least one SE train operated by the Indian Railways.

Based on the data from the train timetable, the average number of SE trains that stop per station was 30 [50]. The historical data on arrival and departure were gathered from Cleartrip.com for the years analyzed. The data also contained information on delays of all the SE trains used to compute travel times between stations. The impedance needed for accessibility calculations was computed by adding the usual train travel time between stations and arrival/departure delays of the trains for 2016–2020. Thus, both the delay data and travel times in the timetables were used. There was an average delay of almost 3 h per day per zone of the Indian Railways for the years 2016 to 2020. The impedance thus obtained was used in the accessibility calculation for each station. The network with stations and rail tracks was used in connectivity calculations. The arrival and departure times were corroborated by various websites and sources that record Indian rail delays at stations, including the National Train Enquiry System of the Indian Railways [58].

Further, each station was mapped to a city it served. The city population data were aggregated to the spatial scale of the zone by extrapolating the city population from the year 2011 population data obtained from the Census Bureau of India [59]. The population data required for calculating the accessibility of stations (and zones) for the five years 2016 to 2020 were obtained by assuming an average population growth of 1% per year in India from the World Bank [60].

The zones of the Indian Railways were constructed on a geographical information system (GIS) software with the railway zone map from MapsofIndia [61], and the spatial map thus produced corroborated with the official webpage of various zones from the Indian Railways [57].

Equations (1) and (2) were utilized to calculate connectivity and accessibility for each station, respectively. Subsequently, the resulting values for each station were normalized against the highest connectivity and accessibility value among all stations across all zones. The average of these normalized values was then computed for the stations within each zone.

It was observed that the total connectivity of stations for the 7 a.m.–9 p.m. period is the largest for the Southern Railway zone. For the 9 p.m.–7 a.m. period, the Northern Railway has the largest total connectivity of the stations. The Northern Railway also has the largest total accessibility for the stations during the 7 a.m.–9 p.m. period, while the South Central Railway has the largest total accessibility for the stations during the 9 p.m.–7 a.m. period.

The values for connectivity, accessibility, and disparity remained relatively consistent across the zones in previous years. For instance, in 2020, as shown in Figure 4, the West Central Railway zone had the highest normalized connectivity during the 7 a.m.–9 p.m. timeframe, while the North Central Railway had the highest normalized connectivity during the 9 p.m.–7 a.m. timeframe. The Northern Railway zone had the greatest accessibility during the 7 a.m.–9 p.m. period, and the Eastern Railway zone led in accessibility during the 9 p.m.–7 a.m. period, as illustrated in Figure 5 for 2020.

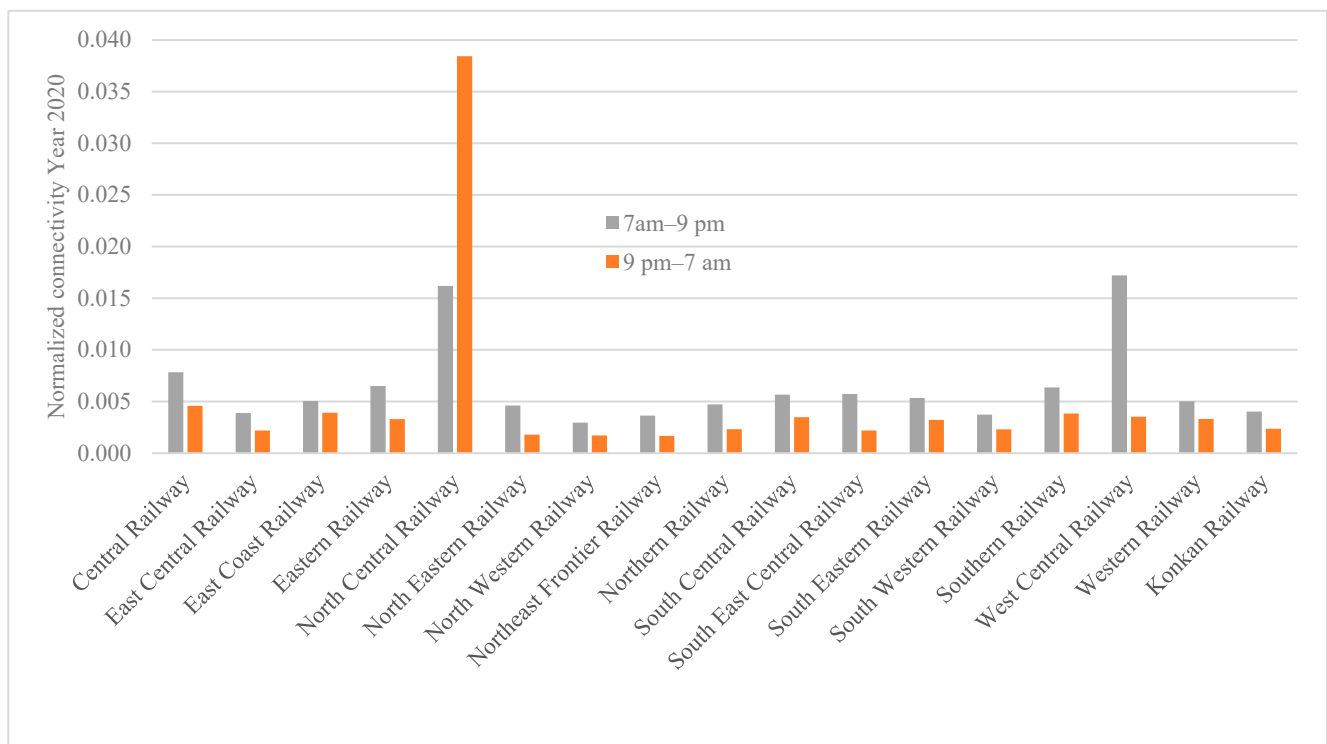


Figure 4. Year 2020 normalized connectivity for zones.

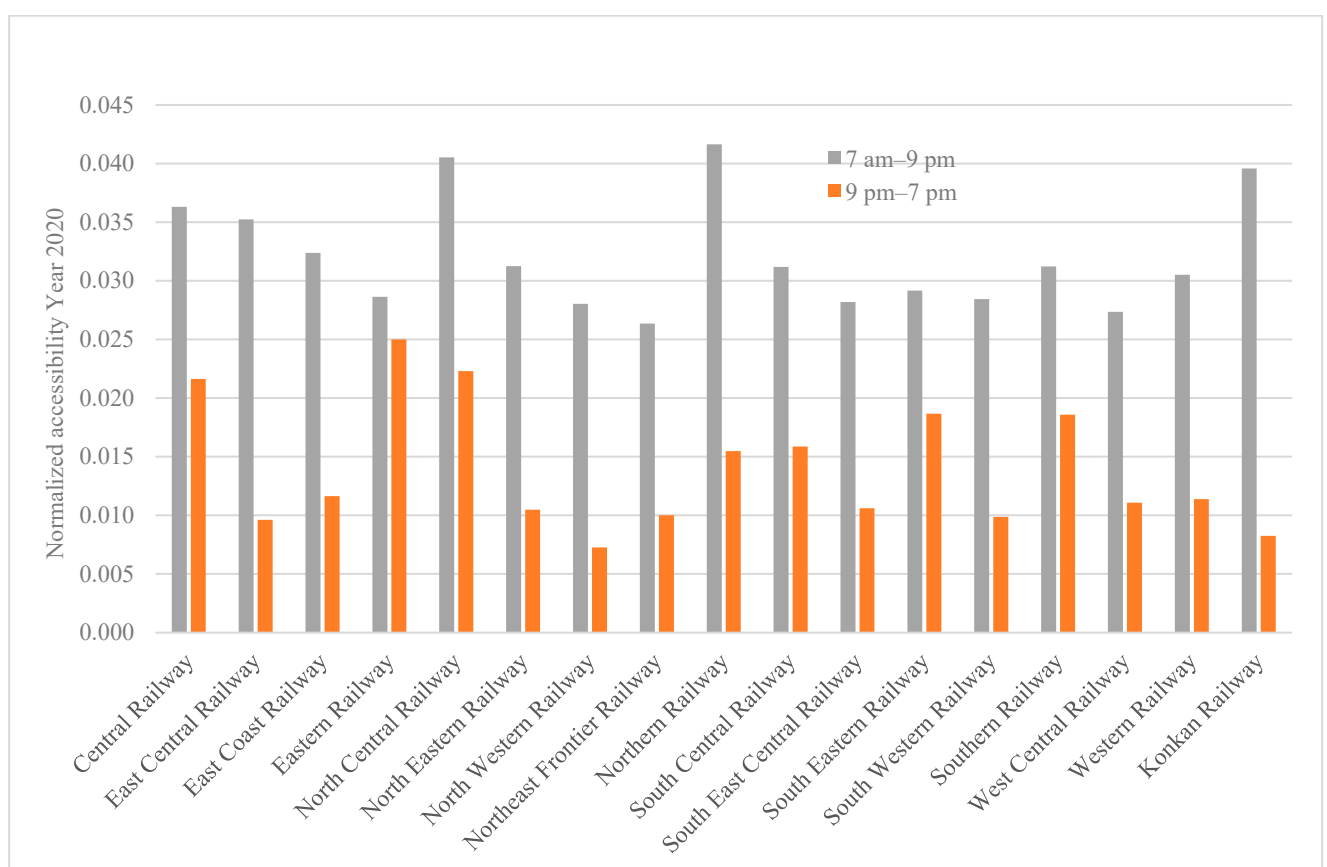


Figure 5. Year 2020 normalized accessibility for zones.

Theil's T index, previously discussed, was used to measure inequalities in both connectivity and accessibility. For example, in 2020, as shown in Figure 6, the South Central Railway zone exhibited the highest connectivity-based inequality in both the 7 a.m.–9 p.m. and 9 p.m.–7 a.m. periods, while the Southeast Central Railway zone had the highest accessibility-based inequality across both timeframes.

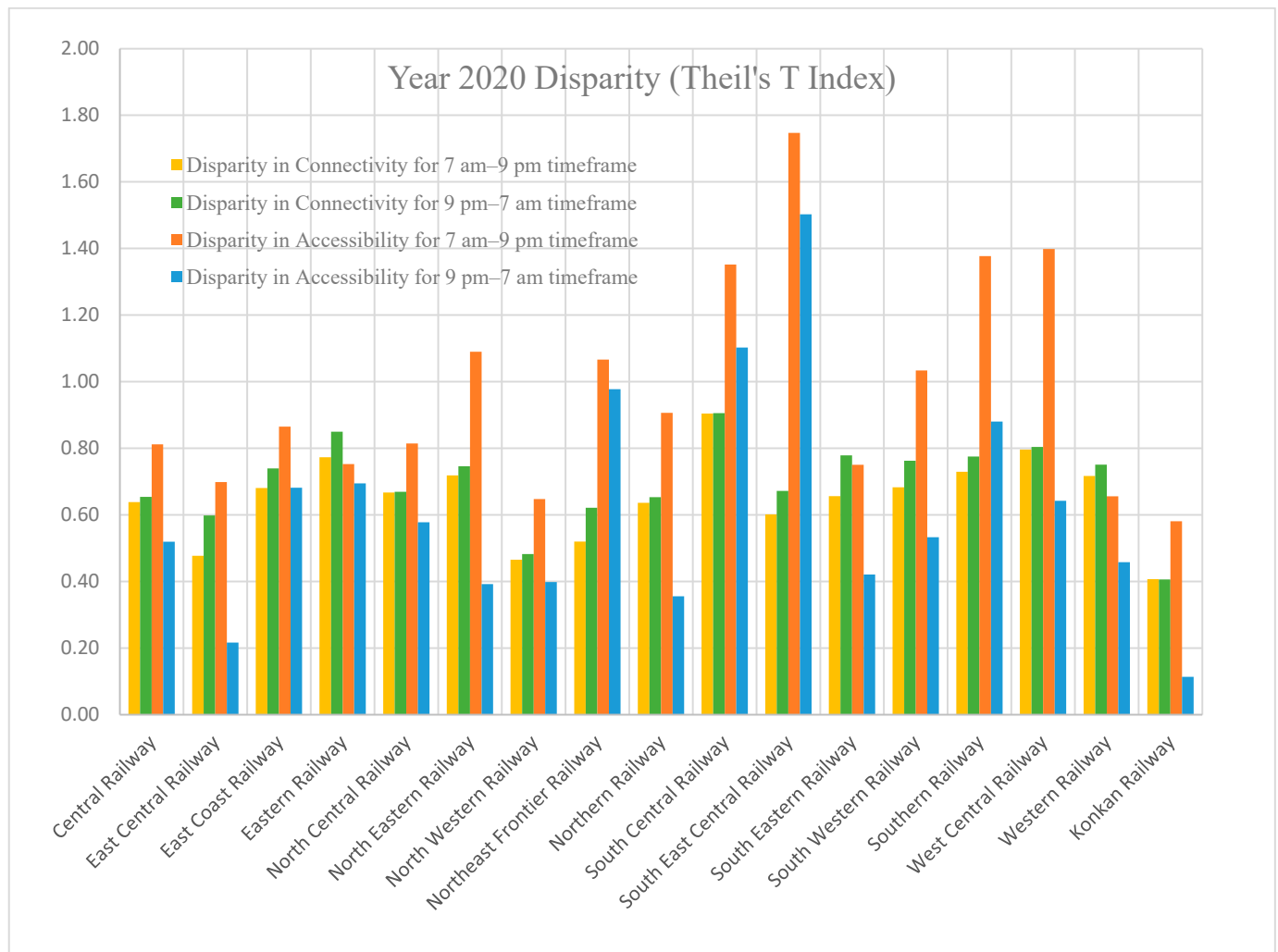


Figure 6. Year 2020 disparity with connectivity and accessibility measures.

The Indian Railways operates in cities with diverse economic backgrounds, each receiving varying levels of investment in their respective zones [31]. These cities, with their associated stations, are unevenly served by the SE trains of the Indian Railways. Figure 7 illustrates the fluctuating investment levels across zones from 2016 to 2020, consistently increasing within specific zones (data source: Indian Railways [62]). The absence of data for the Konkan Railway zone excludes its past investment information from the chart.

According to data from the Indian Railway budget, investments in each zone correlated with the passenger rail revenue generated during the same period [62]. Thus, we assumed that an increase in investment brought about an immediate increase in passenger revenue.

As per the Indian Railways publicly available financial data through budget reports, there is no distinct investment noted for the SE trains. The same rail network infrastructure, such as tracks, signal systems, etc., is used by both the SE trains and other mail/express trains of the Indian Railways. Financial reports of Indian Railways indicate that these investments primarily contribute to total working expenses, encompassing repairs and

maintenance of permanent way and works, carriages and wagons, and operating expenses such as rolling stock, equipment, traffic, and fuel [62].

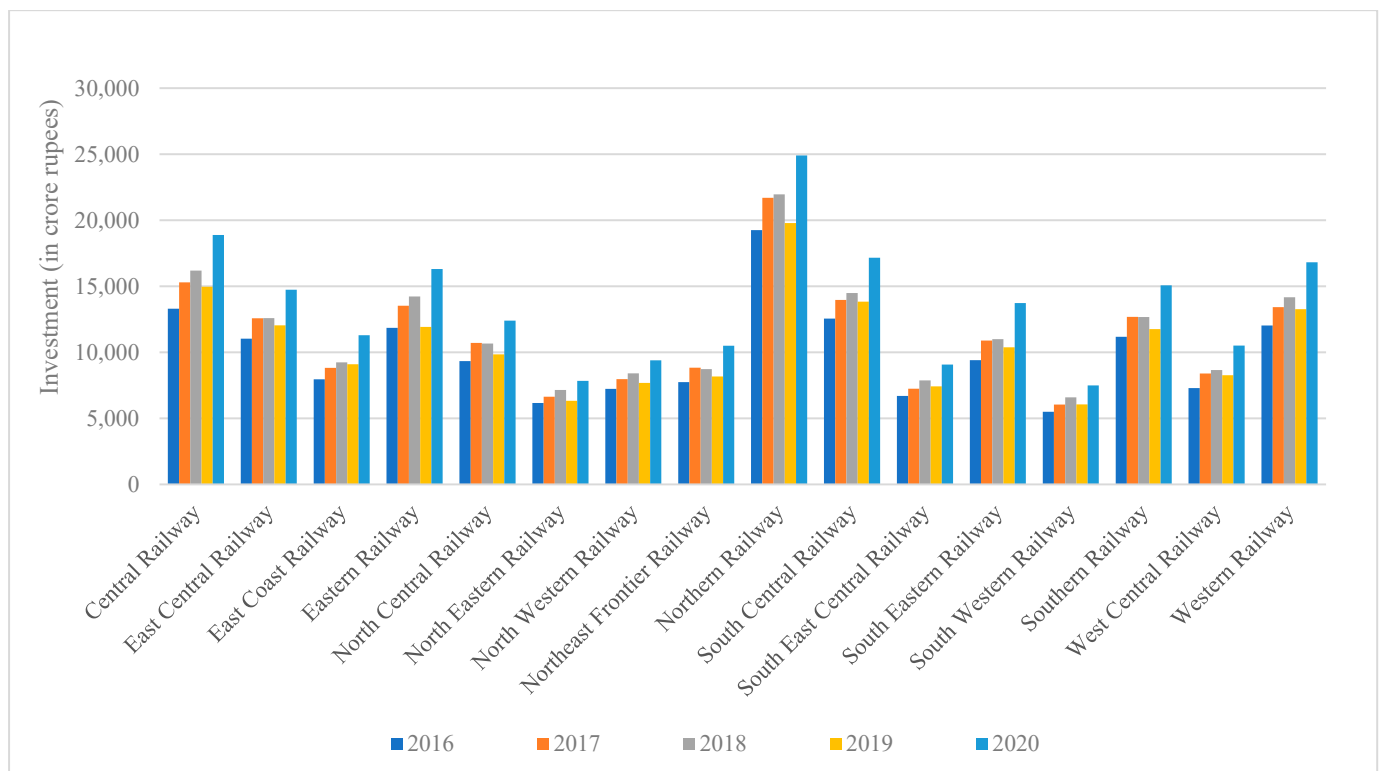


Figure 7. Zonal investment by the Indian Railways during 2016–2020.

5. Results and Discussion

We use the inequality-to-investment ratio defined earlier in Equation (5), supplemented by the evaluation rule of Table 1 presented in the earlier section, to determine zones as winners and losers.

Zones as winners and losers were determined across four analysis periods: 2016–2017, 2017–2018, 2018–2019, and 2019–2020. For the data on the sixteen zones analyzed for five years, eighty data points were obtained if considering connectivity or accessibility for each of the two timeframes, 7 a.m.–9 p.m. and 9 p.m.–7 a.m. The findings are summarized in Figure 8, with the values of the inequality-to-investment ratio presented in each cell.

Please note that in Figure 8, for simplification, the following year ranges are used: 2016–2017 represents 2016–2017, 2017–2018 represents 2017–2018, 2018–2019 represents 2018–2019, and 2019–2020 represents 2019–2020.

The analysis of the connectivity case for the 7 a.m.–9 p.m. timeframe showed that most of the zones were observed to be losers across the four analyzed periods when assessed for connectivity. The South East Central zone was a loser zone, except in 2018–2019, where it was neutral.

For the travel timeframe of 9 p.m.–7 a.m. for connectivity, most of the zones were also noted to be losers, and the number of zones as losers was the highest for the 2016–2017 period. The Northeast Frontier Railway and South East Central Railway zones were considered losers across all four periods.

Accessibility for the 7 a.m.–9 p.m. timeframe showed that almost all of the zones (except North Central, North Western, Northern, and Western) were losers during the 2016–2017 period and neutral during 2018–2019, and the majority of the zones were winners during the periods of 2017–2018 and 2019–2020.

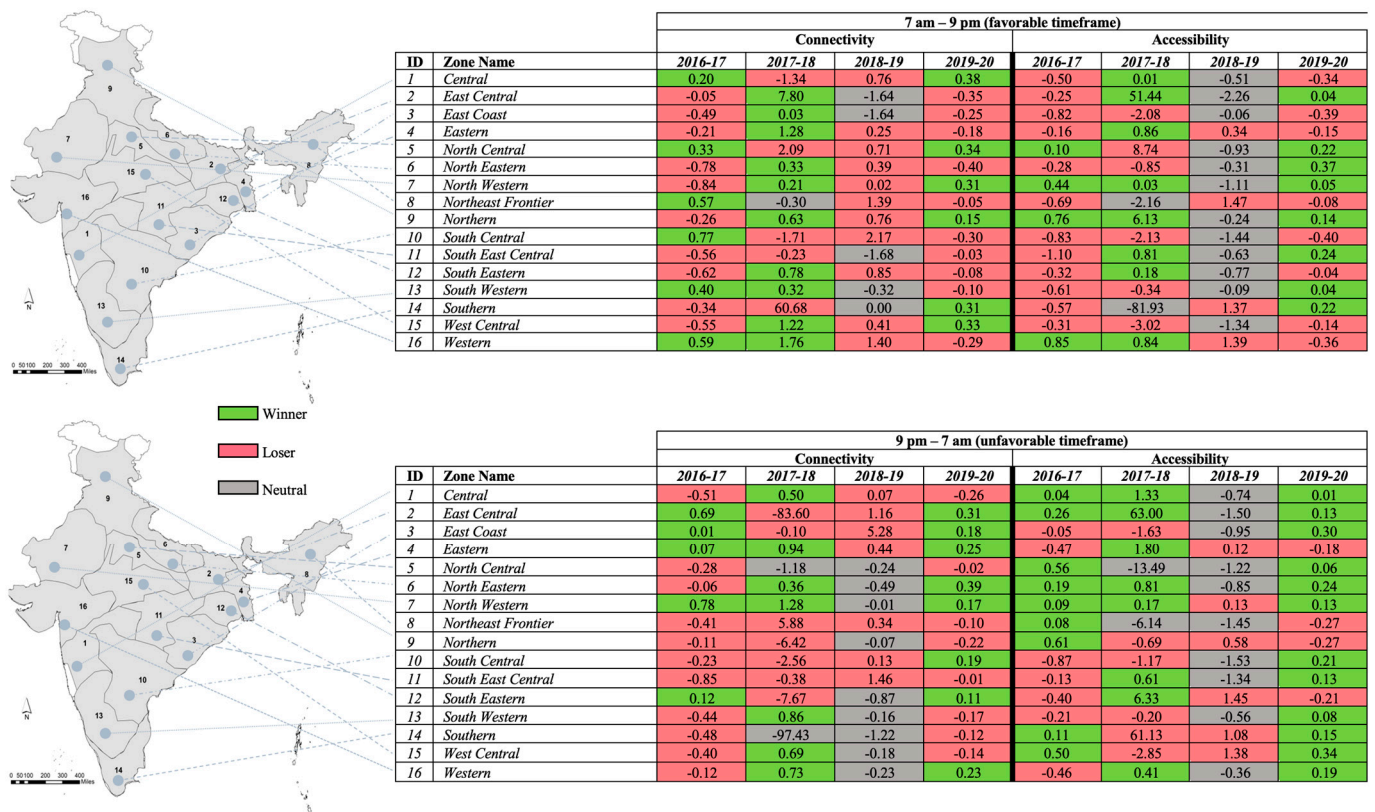


Figure 8. Zones as winner, loser, or neutral across the four periods for the two timeframes (value in cell is the inequality-to-investment ratio).

For accessibility during the 9 p.m.–7 a.m. timeframe, the majority of the zones were winners for the 2019–2020 period. Almost half the zones were winners or losers for the 2016–2017 and 2017–2018 periods. While zones are either neutral or losers for the 2018–2019 period, almost all the zones except Eastern, Northeast Frontier, South Central, and South Western were winners for the 2019–2020 period.

In summary, the findings indicate that most zones experienced an increase in inequality over this timeframe for connectivity, year after year. It can also be interpreted that the zone's connectivity, measured by the number of trains and their delays during the two timeframes, was insufficient to alleviate inequality throughout the analyzed periods.

Overall, for both connectivity and accessibility for 2018–2019 and for the two travel timeframes, none of the zones were winners. This indicates that an increase in investment over the years from 2016 to 2020 did not decrease inequality from 2018 to 2019, though other factors might have influenced this inequality increase that needs further investigation.

Comparing connectivity and accessibility, for the 7 a.m.–9 p.m. timeframe, the highest number of total ten zones repeat status as combined winners, losers, and neutral for each 2016–2017 and 2019–2020 periods. This is evident from the compilation shown in Table 8. For the same timeframe, the highest total of six zones continued as winners from the 2017–2018 period, and eight zones were repeat losers from the 2016–2017 period.

The largest number of zones with repeat status was twelve for the two timeframes (7 a.m.–9 p.m. and 9 p.m.–7 a.m.) for the accessibility measure for the period 2017–2018.

Based on the compilations shown in Table 8, it is evident that there is inconsistency in the number of zones as winners, losers, or neutral across the timeframes, periods, and the two measures. Similar to the year ranges in Figure 8, both Table 8 and the charts in Figure 9 use the following year ranges: 2016–2017 represents 2016–2017, 2017–2018 represents 2017–2018, 2018–2019 represents 2018–2019, and 2019–2020 represents 2019–2020.

Table 8. Frequency of zones repeating their status as winners, losers, or neutral across periods.

		Connectivity (7 a.m.–9 p.m.)				Accessibility (7 a.m.–9 p.m.)				Connectivity (9 p.m.–7 a.m.)				Accessibility (9 p.m.–7 a.m.)			
		2016– 2017	2017– 2018	2018– 2019	2019– 2020	2016– 2017	2017– 2018	2018– 2019	2019– 2020	2016– 2017	2017– 2018	2018– 2019	2019– 2020	2016– 2017	2017– 2018	2018– 2019	2019– 2020
Connectivity (7 a.m.–9 p.m.)	2016–2017	10(2,8,0)				5(0,5,0)											
	2017–2018	9(6,2,1)				8(6,2,0)											
	2018–2019	7(0,3,4)				6(0,4,2)											
	2019–2020	10(4,6,0)				4(1,3,0)											
Accessibility (7 a.m.–9 p.m.)	2016–2017	10(2,8,0)	9(3,6,0)														
	2017–2018	9(6,2,1)	12(7,4,1)														
	2018–2019	7(0,3,4)	10(0,2,8)														
	2019–2020	10(4,6,0)	10(7,3,0)														
Connectivity (9 p.m.–7 a.m.)	2016–2017	5 (0,5,0)	6(2,4,0)														
	2017–2018	8(6,2,0)	9(5,3,1)														
	2018–2019	6(0,4,2)	5(0,1,4)														
	2019–2020	4(1,3,0)	8(6,2,0)														
Accessibility (9 p.m.–7 a.m.)	2016–2017	9(3,6,0)	6(2,4,0)														
	2017–2018	12(7,4,1)	9(5,3,1)														
	2018–2019	10(0,2,8)	5(0,1,4)														
	2019–2020	10(7,3,0)	8(6,2,0)														

a(b,c,d) stands for: a = number of zones repeating status. b = number of zones repeating status as winner zones. c = number of zones repeating status as loser zones. d = number of zones repeating status as neutral zones.

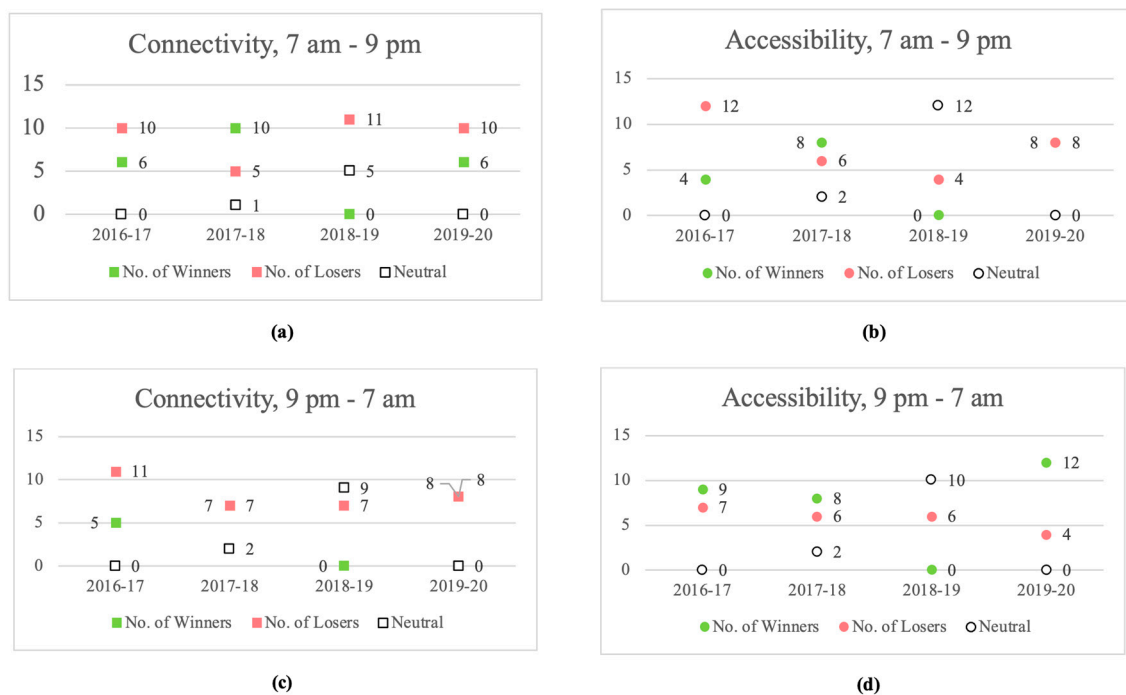


Figure 9. Count of zones as winners and losers (a) connectivity, 7 a.m.–9 p.m. (b) accessibility, 7 a.m.–9 p.m. (c) connectivity, 9 p.m.–7 a.m. (d) accessibility, 9 p.m.–7 a.m.

Essentially, the comprehensive analysis of the results presented in Table 8 yields two crucial insights. Firstly, the count of zones identified as repeat winners, losers, or neutral varies depending on the measures (be it connectivity or accessibility) and the considered timeframes. Secondly, none of the four periods consistently exhibits the same distribution of winners, losers, or neutral zones across both measures and timeframes. This variability underscores the disparities among zones and highlights the complexity involved in deducing guidance for investments, resource allocation, and the introduction or scheduling of new SE trains.

For both connectivity and accessibility, the number of loser zones was higher than the number of winners for the 7 a.m.–9 p.m. timeframe for 2016–2017. See Figure 9. On the other hand, during this same travel timeframe, the number of zones as winners was higher than the number of loser zones during 2017–2018 for connectivity and accessibility.

It was found for the 9 p.m.–7 a.m. timeframe for connectivity that the number of loser zones was higher than the number of winner zones compared to accessibility; this was observed for 2016–2017, 2017–2018, and 2019–2020. Therefore, if accessibility was considered the basis of analysis, the disparity among zones seemed to have reduced, possibly due to service improvements of SE trains. This could have occurred due to their reduced delays or other means of improving efficiency and reliability through upgrades in the rail infrastructure with investments by the Indian Railways for the respective zones.

An aggregate level of analysis was conducted using the odds ratio, signifying the performance persistency of zones as winners and losers from one analyzed period to another. A ratio greater than 1 meant that a zone that was a loser in one period had remained a loser during the next period with a 50% probability, thus promoting inequality. On the contrary, the ratios below 1 meant that a zone that was a winner in a period was a loser in the next period with a probability above 50%, which also promoted inequality.

With reference to the information above, odd ratios are computed and compiled as shown in Tables 9 and 10. It was observed that for the 9 p.m.–7 a.m. timeframe for connectivity, a zone that was a loser zone continued to be a loser zone for the periods 2017–2018 to 2018–2019 and 2018–2019 to 2019–2020 for the 7 a.m.–9 p.m. timeframe. For accessibility, however, zones that were losers continued to be losers for all the periods when

considering the 9 p.m.–7 a.m. timeframe. Therefore, these findings indicate that disparities among zones continued with a 50% probability when considering the number of loser zones presented in Figure 8, in which case, for accessibility, a significant number of loser zones continued to be the loser from 2016 to 2020 during 9 p.m.–7 a.m. timeframe.

Table 9. Cross-product ratios for connectivity.

Timeframe	2016–2017 to 2017–2018	2017–2018 to 2018–2019	2018–2019 to 2019–2020
9 p.m.–7 a.m.	0.58	2.20	0.78
7 a.m.–9 p.m.	0.21	0.47	1.18

Table 10. Cross-product ratios for accessibility.

Timeframe	2016–2017 to 2017–2018	2017–2018 to 2018–2019	2018–2019 to 2019–2020
9 p.m.–7 a.m.	1.11	1.01	1.2
7 a.m.–9 p.m.	2.33	0.20	2.33

Exclusively for the observation from the period 2016–2017 to 2017–2018, the ratios were less than 1 for connectivity and higher than 1 for accessibility. If connectivity was used to assess inequality, a zone that is a loser had transformed into a winner zone from 2016–2017 to 2017–2018 with a 50% probability, thereby improving inequality. On the other hand, with an assessment based on accessibility, a zone had remained a loser zone from 2016–2017 to 2017–2018, aggravating the disparity. Thus, the choice of the performance measure, whether connectivity or accessibility, matters in determining the success of policy-related interventions (such as investments) in improving zone disparities.

6. Conclusions

In this research, a framework for evaluating inequality is presented for investment received for zones within which passenger rails operate. The inequality is advertently created when trains stop during unfavorable timeframes, dictated by the trains' arrivals and departures schedules, such as during nighttime when the station lacks first-/last-mile connectivity or safety considerations around the station are important. Using an inequality-to-investment ratio developed, winners and losers over a given period of time are determined.

A case study was carried out with the superfast express (SE) trains of the Indian Railways for the favorable (7 a.m.–9 p.m.) and unfavorable (9 p.m.–7 a.m.) timeframes of the passenger travel. It was found that zone investments, when accounting for inequalities, created a variable number of zones as winners, losers, or neutral across the timeframes, evaluation periods, and the two measures: connectivity and accessibility. All the zones were either losers or neutrals during the 2018–2019 period for either of the two timeframes and for connectivity and accessibility measures. Specific outputs showed that for accessibility, the 9 p.m.–7 a.m. timeframe saw more winners than losers during both the 2016–2017 and 2019–2020 periods. A very high number of zones were losers than winners for accessibility during the 2016–2017 period for the 7 a.m.–9 p.m. period.

Therefore, for the investments that did not diminish the inequality of a zone and inadvertently created a loser zone, it can be inferred that investments must be increased or reconsidered to increase connectivity and accessibility for specific zones.

It is further suggested that any possible schedule changes in the SE train in the Indian Railways zones that are marginalized (less strategically served) be made to address their connectivity and accessibility during 'unfavorable' travel hours of 9 p.m.–7 a.m. Furthermore, appropriate investments and provisions for first-/last-mile transport around stations during unfavorable timeframes should be made for zones to reduce their inequalities compared to other zones. Otherwise, the zones already receiving significant investment

and being well connected and accessible with passenger rails will continue to experience economic prosperity. In contrast, zones with SE train arrivals and departure schedules mostly during ‘unfavorable’ times might not experience the desired economic growth by SE trains connecting to India’s megacities.

The case study results and interpretations of the outcome are mainly applicable to the zones of the Indian Railways analyzed, as the investment data were only available and utilized at this spatial scale. However, the framework presented in this research is generic because of its simple approach and could be applied to evaluate inequalities for any entity, whether city, state, or region receiving investment or government-led funds. Therefore, using the inequality evaluation framework presented in this paper to identify the winning and losing entities through the simplified approach proposed could be very useful. It would provide first-hand information to agencies for infrastructure development, scheduling, and resource allocation, ultimately creating a more balanced and sustainable railway system.

Our future research will focus on applying the framework outlined in this paper to evaluate the inequality across the entire Indian passenger rail mobility system, encompassing SE trains and all other mail/express trains operated by Indian Railways.

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